

A MANUFACTURED APÉRY APPARATUS FOR A CUSP-FORM L -VALUE: FRICKE-ODD SOURCES, FOLD REGULARIZATION, AND THE LIMIT $L(f_6, 3)/2$

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ABSTRACT. Every Apéry-like recurrence in the sporadic class carries a canonical *companion source* Φ , a weight- $(r+1)$ modular form whose Eichler integral is the second solution. For all twelve identifiable sporadic families Φ is purely Eisenstein, which is why the classical Apéry limits are zeta and Dirichlet L -values. This paper asks the design question in the other direction: can one *manufacture* a recurrence, with Apéry-quality integrality and exponential convergence, whose limit is a critical L -value of a *cusp form*? We answer yes, and exhibit the apparatus. On Domb’s curve (level 12) the Fricke-odd combination $f^* = f_6(q) - 4f_6(q^2)$ of the level-6 newform $f_6 = (\eta_1\eta_2\eta_3\eta_6)^2$ vanishes at the Atkin–Lehner fixed point — we prove this by an oldspace eigenvector argument — and satisfies $f^*\sqrt{1-4t} = \Phi_\alpha$. The resulting companion B^* obeys the Domb recurrence with central-binomial forcing, has $d_n^3 B_n^* \in \mathbb{Z}$ [VERIFIED: $n \leq 40$], converges at rate 4^{-n} , and [VERIFIED: 61-digit PSLQ] satisfies $B_n^*/A_n \rightarrow L(f_6, 3)/2$: a bare critical L -value, with no quasiperiod and no $\pi L(f_6, 2)$ term. The mechanism is a single parity: Fricke-oddness both regularizes the fold (exponential convergence) and kills the even critical value through $\Lambda^*(2) = 0$. We contrast this with a level-6 experiment in which the Fricke sign is $+1$ and the limit is contaminated by an unidentified second-kind constant [OPEN]. A δ -control run and a classification corollary show that α is the *unique* sporadic family admitting the construction. Finally, we do the irrationality bookkeeping honestly: 4^n against $d_n^3 \sim e^{3n}$ loses, so no irrationality follows. Every statement carries its evidence label; finite computations are never reported as proofs.

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1. INTRODUCTION

1.1. **Apéry’s pair, and what a “companion source” is.** Apéry’s proof that $\zeta(3) \notin \mathbb{Q}$ runs on a pair of solutions of one recurrence:

$$m^3 y_m = (2m-1)(17(m-1)^2 + 17(m-1) + 5) y_{m-1} - (m-1)^3 y_{m-2}, \quad (1)$$

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the integral solution $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and a second solution B_n with $B_n/A_n \rightarrow \zeta(3)$, whose denominators are controlled by d_n^3 , $d_n = \text{lcm}(1, \dots, n)$. The irrationality follows because the error $|A_n \zeta(3) - B_n|$ decays like α^{-n} with $\alpha = (1 + \sqrt{2})^4 = 33.97\dots$, faster than the denominators $d_n^3 \sim e^{3n} = 20.08^n$ grow.

The companion paper [1] proves the structural statement behind this pair for all fifteen sporadic Apéry-like families. Write r for the order of the generating differential operator ($r = 3$ for the six “weight-3” families, including Apéry’s $\zeta(3)$ family γ), q for the nome of the operator, $t(q)$ for the inverse map, $F(q)$ for the analytic solution pulled back to q , $\sigma = \theta_t \log q$, and P for the leading polynomial of the recurrence. Then the normalized companion is an Eichler-type integral,

$$y_B = F \cdot \theta_q^{-r} \Phi, \quad \Phi := \frac{t \sigma^r}{P(t) F}, \quad (2)$$

valid for all n [PROVED] [1, Thm. 3.4]. We call Φ the *companion source*. Once the family’s modular parametrization is identified, Φ is a q -series of modular weight $r + 1$ on the family’s level, and the Apéry limit is the value of its Eichler integral at the “fold” — the branch point of $t(q)$ nearest the origin.

1.2. The Eisenstein/cuspidal dichotomy. The programme’s source ledger [2] settles what these Φ are. For all twelve sporadic families whose parametrization is identified — the six order-2 families A – F and the order-3 families $\alpha, \gamma, \delta, \varepsilon, \zeta, \eta$ — the source Φ is a *pure Eisenstein series* of weight $r + 1$, with an explicit divisor-sum coefficient law, with *no cuspidal component anywhere* [VERIFIED: exact $\mathbb{Q}$ -arithmetic coefficientwise to q^{60} , with the band q^{27} – q^{60} held out from the fit]. (The three remaining Cooper rows are coordinate-obstructed and assert no modular weight.) This single fact explains the entire limit column of the sporadic table: an Eisenstein source has an Eichler integral built from Lerch-type sums, so the fold value is a zeta value or a Dirichlet L -value, and nothing else can occur. In particular γ reproduces Beukers’s weight-4 level-6 Eisenstein picture for $\zeta(3)$, and D reproduces the $\Gamma_1(5)$ weight-3 picture for $\zeta(2)$, both recovered here *from the recurrence alone*.

1.3. The design question. The dichotomy immediately poses the question this paper answers.

Problem 1.1 (the manufacturing question). Can one *manufacture* a recurrence — integral, three-term with an integral forcing term, with d_n^r -quality denominators and exponentially small error — whose Apéry limit is a *critical L -value of a cusp form*?

Nothing in the sporadic table can answer it by inspection: the table’s sources are Eisenstein by [2], so no sporadic limit is cuspidal. The point of (2), however, is that the source is an *input*. Given a rectified family, *any* modular form of the right weight and level may be substituted for Φ in (2); the resulting series still satisfies the family’s recurrence, now with an inhomogeneity determined by the substituted form. Manufacturing an apparatus therefore means choosing (level, family, source) so that three separate things go right at once: the inhomogeneity must be integral (or the recurrence is not arithmetic), the error must decay exponentially (or the limit is useless), and the fold value must be a clean L -value rather than a contaminated mixture.

We show these three requirements are governed by three distinct structures — the d_n^r integral structure of the family, the *vanishing* of the source at the fold, and the *parity* of the source under the Atkin–Lehner (Fricke) involution — and we exhibit a triple where all three succeed simultaneously.

1.4. Results.

- (1) **A level-6 experiment** (§3). Substituting the unique newform $f_6 = (\eta_1 \eta_2 \eta_3 \eta_6)^2 \in S_4(\Gamma_0(6))$ into Apéry’s own $\zeta(3)$ curve gives the identity $f_6 \sqrt{1 - 34t + t^2} = \Phi_\gamma$ [VERIFIED: q^{60} ; Sturm bound 8], hence Apéry’s recurrence (1) with the integral forcing term $w_n = [t^n](1 - 34t +$

$t^2)^{-1/2} = 1, 17, 433, 12257, \dots$, and $d_n^3 B_n^f \in \mathbb{Z}$ [VERIFIED: $n \leq 40$]: *the cusp form costs nothing in denominators*. The limit decomposes as $\Theta(\tau_c) = L(f_6, 3) - \frac{\pi}{\sqrt{6}} L(f_6, 2)$ [VERIFIED: 120 digits] plus a second-kind term $\frac{2\pi}{\sqrt{6}} \Theta_2(\tau_c)$ which PSLQ cannot reduce [OPEN]. Convergence is only polynomial. This experiment is the negative that fixes the design.

- (2) **The construction** (§4). At level 12, $\dim S_4(\Gamma_0(12)) = 2$; we prove (Theorem 4.2) that W_{12} acts on the oldspace by $f_6(q) \mapsto 4f_6(q^2)$, so the odd eigenvector $f^* = f_6(q) - 4f_6(q^2)$ vanishes at the fixed point $\tau_{12} = i/\sqrt{12}$, which is exactly the fold of the Domb family α . The source identity is $f^* \sqrt{1-4t} = \Phi_\alpha$ [VERIFIED: exact to q^{40}], giving the Domb recurrence with central-binomial forcing, $d_n^3 B_n^* \in \mathbb{Z}$ [VERIFIED: $n \leq 40$], error $\sim 4^{-n}$, and

$$\boxed{\lim_{n \rightarrow \infty} \frac{B_n^*}{A_n} = \frac{L(f_6, 3)}{2}} \quad [\text{VERIFIED: 61-digit PSLQ, residual } 2 \times 10^{-61}].$$

- (3) **The mechanism** (§5): one parity, two effects. Fold-vanishing regularizes the inhomogeneity (exponential convergence); Fricke-oddness forces $\Lambda^*(2) = 0$, killing the even critical value, so a single critical value survives. The Fricke functional equation is [PROVED (route complete)]; the exact rational factor $\frac{1}{2}$ is [ROUTE WRITTEN, CONSTANT OPEN].
- (4) **Controls and scope** (§6): the δ control isolates fold-vanishing as the operative hypothesis, and a classification corollary shows α is the *unique* order-3 sporadic family admitting the construction.
- (5) **Honest bookkeeping** (§7): 4^n versus $d_n^3 \sim e^{3n}$ loses, so no irrationality statement is made or implied. We record the exact scan over all fifteen families that locates precisely Apéry's two classical wins and nothing else, and the discovery that Apéry's own source vanishes at the fold — i.e. his miracle is the *same* construction in the Eisenstein class.

1.5. **Credit.** The programme of substituting cuspidal sources, the six-item theorem package that organizes §4–§5, the refined hypotheses of Conjecture 6.4, the directed δ control of §6.1, and the irrationality-race scoring of §7 are all due to the collaborator Sol, in a sequence of project slates and adversarial reviews. Where a statement is Sol's, it is attributed at the point of use.

1.6. **Evidence discipline.** [PROVED]/[THM]: complete proof (computer algebra only for finite exact rational identities). [PROVED (route complete)]: a complete mathematical route with all steps written, but not yet typeset as a referee-ready proof; nothing in this class is used as a hypothesis of a [PROVED] statement. [VERIFIED: range]: exact or high-precision computation over a stated finite range — *never* a proof. [EXCLUDED]: negative established by proof, or by exact computation over a precisely stated search space (the space is always stated). [OPEN]: open. [ROUTE WRITTEN, CONSTANT OPEN]: the structural derivation is complete but a normalizing constant is not certified. Two standing rules: finite coefficient agreement is never called an identity unless a Sturm bound converts it, and finite integrality is never called modularity or arithmeticity.

2. SET-UP: SOURCES, FOLDS, AND APÉRY LIMITS

This section fixes notation and records, without reproving, the base results of [1, 2] that we use.

2.1. **Rectification and the companion.** Let L be an order- r operator in θ_t over $\mathbb{Q}(t)$ with maximal unipotent monodromy at $t = 0$ and Frobenius basis $y_0 = 1 + O(t)$, $y_1 = y_0 \log t + g, \dots$. The nome is $q = t \exp(g/y_0)$, and L is *exactly rectified* if $L = C(t) F \theta_q^r F^{-1}$ for a rational C ; equivalently the kernel of L is $\text{span}\{F, F \log q, \dots\}$. All fifteen sporadic operators are exactly rectified [PROVED] [1, Thm. 3.6, Cor. 3.7], and then (2) holds for all n [1, Thm. 3.8].

Definition 2.1 (substituted companion). Let $\Psi(q) \in q\mathbb{Q}[[q]]$ be any series. The Ψ -companion of the family is

$$y^\Psi := F \cdot \theta_q^{-r} \Psi, \quad B_n^\Psi := [t^n] y^\Psi.$$

By rectification, $Ly^\Psi = C(t)\Psi \circ q(t)$; writing $R(t) := t\Psi/\Phi$ (a series in t , since Φ and Ψ both begin at q), the sequence B^Ψ satisfies the family's homogeneous recurrence with forcing term $[t^n]R$. Taking $\Psi = \Phi$ gives $R = t$ and recovers the canonical companion.

Thus *the choice of source is a free design parameter*, and the entire arithmetic of the manufactured apparatus is the arithmetic of the ratio $R = t\Psi/\Phi$ together with the family's own integral structure.

2.2. The fold and the connection value. The map $q \mapsto t(q)$ is 2-to-1 near a simple branch point q_c (the *fold*): $t - t_c \simeq c(q - q_c)^2$. For the families we treat, t_c is the smaller root of $P(t)$, and $A_n \sim t_c^{-n} \cdot (\text{subexp})$.

Lemma 2.2 (fold connection). [PROVED] *Let y_0, y_B be analytic in q near q_c and let t have a simple fold there. Then $\lim_{n \rightarrow \infty} B_n/A_n = y'_B(q_c)/y'_0(q_c)$, provided the limit exists.*

Proof. Decompose y_0, y_B into even and odd parts under the local fold involution $q \mapsto \iota(q)$ fixing t . The even parts are analytic functions of t at t_c ; the odd parts carry the square-root branch, so the singular part of each y as a function of t at $t = t_c$ is (odd part), itself of the form $(q - q_c) \cdot (\text{unit})$ times $\sqrt{t - t_c}$ -data. The asymptotics of $[t^n]y$ are governed by the singular part; the ratio of the two singular parts is the ratio of the odd parts, i.e. of the q -derivatives at q_c . Analytic (even) parts contribute exponentially smaller terms. $\square$

Written out, for a source Ψ with Eichler integral $\Theta_\Psi := \theta_q^{-r} \Psi$,

$$\xi^\Psi = \Theta_\Psi(q_c) + \frac{F(q_c)}{F'(q_c)} \Theta'_\Psi(q_c), \quad (3)$$

which is the formula used throughout: a *period-type* term plus a *quasiperiod-type* term. The second term is where the difficulty lives.

Remark 2.3 (the level of rigour of the modular inputs). Statements like “ Φ_γ is the weight-4 level-6 Eisenstein series with coefficient law $\sigma_3(m) - 28\sigma_3(m/2) + 63\sigma_3(m/3) - 36\sigma_3(m/6)$ ” are theorems *conditional* on the certification that the η -quotient expressions for t and F are modular of the stated weight, level and character and satisfy the family ODE. That certification is complete for the level-9 family ζ and routine but unwritten for the others [2]. We flag this once here and do not repeat it; it affects the *interpretation* of the identities below, not their exactness as q -series statements.

3. THE LEVEL-6 EXPERIMENT: A CUSPIDAL SOURCE ON APÉRY'S OWN CURVE

The first thing to try is the cheapest thing: keep Apéry's curve and swap the source. This experiment was proposed by Sol (Project D) and is instructive precisely because it half-fails.

3.1. The substitution. Family γ has $r = 3$, $P_\gamma(t) = 1 - 34t + t^2$, A_n the Apéry numbers, and (as recovered in [2]) an Eisenstein source Φ_γ of weight 4 and level 6. Now $\dim S_4(\Gamma_0(6)) = 1$, spanned by the newform

$$f_6 = (\eta_1 \eta_2 \eta_3 \eta_6)^2 = q - 2q^2 - 3q^3 + 4q^4 + 6q^5 + 6q^6 - 16q^7 + \cdots \quad (\text{LMFDB 6.4.a.a}).$$

We substitute $\Psi = f_6$ in Definition 2.1.

Computer-assisted finite verification 3.1 (the eta identity). [VERIFIED: coefficientwise to q^{60} , exact $\mathbb{Q}$]

$$f_6^2 \cdot P_\gamma(t) = \Phi_\gamma^2, \quad \text{equivalently} \quad f_6 \sqrt{1 - 34t + t^2} = \Phi_\gamma.$$

Both sides lie in $M_8(\Gamma_0(6))$, whose Sturm bound is 8. Hence, conditional on the modularity certification of t, F, Φ_γ , the identity is theorem-grade: agreement to q^{60} vastly exceeds the bound.

The reader new to this style should note the shape of the argument. A finite computation is not a proof; but a finite computation that verifies an identity between two elements of a finite-dimensional space of modular forms, past that space's Sturm bound, *is* converted into a proof by the Sturm criterion. We use this conversion whenever it is available, and say so; where the ambient space is not pinned down we do not.

3.2. The forced recurrence and its integrality. By Definition 2.1,

$$R(t) = t f_6 / \Phi_\gamma = t / \sqrt{1 - 34t + t^2}.$$

The reconstruction is exact: fitting a rational function times a square root against the exact t -series determines $(P, \text{numerator})$ uniquely at degree 2.

Proposition 3.2 (the level-6 forced recurrence). [PROVED] (given Verification 3.1) Put $w_n := [t^n](1 - 34t + t^2)^{-1/2}$. Then $w_0 = 1$, $w_1 = 17$, and

$$(n + 1)w_{n+1} = 17(2n + 1)w_n - n w_{n-1},$$

so $w = 1, 17, 433, 12257, \dots \in \mathbb{Z}$; and $B_n^f := B_n^{f_6}$ satisfies Apéry's recurrence with this forcing:

$$m^3 B_m^f = (2m - 1)(17(m - 1)^2 + 17(m - 1) + 5)B_{m-1}^f - (m - 1)^3 B_{m-2}^f + w_{m-1},$$

with $B_0^f = 0$, $B_1^f = 1$; e.g. $B_2^f = 67/4$, $B_3^f = 12515/36$.

Proof. The recurrence for w is the Gauss contiguous relation for $(1 - 34t + t^2)^{-1/2}$, whose coefficients are Legendre polynomials evaluated at 17; integrality follows by induction from $w_0, w_1 \in \mathbb{Z}$ together with the classical integrality of $P_n(17) \cdot (\text{Legendre normalization})$ — concretely, $w_n = P_n(17)$ and the three-term Legendre recurrence with the stated initial data produces integers because at each step $(n + 1) \mid (17(2n + 1)w_n - n w_{n-1})$, which is the classical statement that $P_n(x) \in \mathbb{Z}[x]$ has integral values at odd integers of this shape. The recurrence for B^f is Definition 2.1 with $R = t/\sqrt{P_\gamma}$: the forcing term is $[t^m]R = w_{m-1}$. $\square$

Computer-assisted finite verification 3.3 (Apéry-quality denominators). [VERIFIED: exact, $n \leq 40$] $d_n^3 B_n^f \in \mathbb{Z}$ for all $n \leq 40$, $d_n = \text{lcm}(1, \dots, n)$.

This is the experiment's clean positive: *the cusp form costs nothing in integrality*. Whatever obstructs a cuspidal irrationality proof, it is not the denominators — at least not at this level, over this range.

3.3. The limit, and the contamination. The fold of γ is the Fricke fixed point $\tau_c = i/\sqrt{6}$, i.e. $q_c = e^{-2\pi/\sqrt{6}}$ [VERIFIED: matches the fold nome to series precision]. Write, for the newform $f_6 = \sum a_n q^n$,

$$\Theta(\tau) := \sum_{n \geq 1} \frac{a_n}{n^3} q^n, \quad \Theta_2(\tau) := \sum_{n \geq 1} \frac{a_n}{n^2} q^n,$$

the third and second Eichler integrals.

Computer-assisted finite verification 3.4 (Eichler value at the Fricke point). [VERIFIED: 120 digits, residual 1.2×10^{-121}]

$$\Theta(\tau_c) = L(f_6, 3) - \frac{\pi}{\sqrt{6}} L(f_6, 2).$$

Proof route [PROVED (route complete)]: the Fricke sign of f_6 is $\varepsilon = +1$ (from $a_2 = -2, a_3 = -3$, giving $\lambda_2 = -a_2/2 = 1, \lambda_3 = -a_3/3 = 1$); evaluate the degree-2 period polynomial of f_6 under W_6 at the fixed point, and use $\Lambda(1) = \Lambda(3)$ to remove $L(f_6, 1)$.

Computer-assisted finite verification 3.5 (the fold value). [VERIFIED: formula versus direct fold computation agree to 2×10^{-29}] The eta transformation gives $F_\gamma(-1/(6\tau)) = -6\tau^2 F_\gamma(\tau)$, whence $F'(\tau_c)/F(\tau_c) = 6\tau_c$ exactly, and (3) becomes

$$\xi_f = \lim_{n \rightarrow \infty} \frac{B_n^f}{A_n} = \Theta(\tau_c) + \frac{2\pi}{\sqrt{6}} \Theta_2(\tau_c) = 0.264718537218080276566224788434303077935 \dots$$

Observation 3.6 (the two defects). (1) *Contamination*. $\Theta_2(\tau_c) = 0.0740870938237972 \dots$ is a *second-kind* Eichler value. PSLQ finds no relation with the critical values $L(f_6, s)$, the derivative values $L'(f_6, 2), L'(f_6, 3), \zeta(3)$, powers of π , or log-twists [EXCLUDED: PSLQ over the stated bases, tolerance 10^{-25} ; all hits spurious]. Status of the constant: [OPEN]. Hence ξ_f is not a bare L -value.

(2) *Slow convergence*. $B_n^f - \xi_f A_n$ decays only polynomially relative to A_n (numerically like $\alpha^n n^{-7/2}$ against $A_n \sim \alpha^n$), so naive ratio extrapolation stalls near 8 digits. The cause is visible in $R = t/\sqrt{P_\gamma}$: the inhomogeneity is *singular at the fold*. Equivalently $f_6(\tau_c) \neq 0$.

Observation 3.7 (why the Eisenstein table looked clean). For Eisenstein sources the second-kind term in (3) is elementary and cancels against the quasiperiod of F — a Legendre-type relation — so the fold value is a bare L -value. The cuspidal experiment shows that this cancellation is a *feature of the Eisenstein class*, not a general mechanism. The general shape is

$$\text{fold value} = \text{period-polynomial part} + \text{quasiperiod} \times \text{second-kind part}.$$

3.4. The problem the experiment leaves.

Problem 3.8 (D-1, Sol). Find a linear form in the family's solutions whose limit is exactly the period part — i.e. kill the second-kind term — and determine its denominator growth. If d_n^3 -quality survives, this is a rational-approximation apparatus aimed directly at a cuspidal L -value.

Two routes present themselves; one is excluded.

Computer-assisted finite verification 3.9 (κ -cancellation excluded). [EXCLUDED: PSLQ, 40 digits, over $\mathbb{Q}$ and over real quadratic fields with stated height caps] The ratio $\kappa = \Theta_{f,2}(\tau_c)/\Theta_{\Phi,2}(\tau_c) = 1.39173177734068830824 \dots$ is neither rational nor quadratic. Hence no $\mathbb{Q}$ -combination of the level-6 cuspidal and Eisenstein companions cancels the quasiperiod term.

That leaves the structural route: *change the source so that the second-kind term is never generated*. By Observation 3.6, Apéry-quality error requires a source *vanishing at the fold*. At level 6 this is impossible: $\dim S_4(\Gamma_0(6)) = 1$, so the only cuspidal direction is f_6 itself, and $f_6(\tau_c) \neq 0$. One must go to a level with a two-dimensional cusp space. This is the whole design.

4. THE CONSTRUCTION: DOMB'S CURVE AT LEVEL 12

4.1. The choice. Take the sporadic family α (Domb's numbers), with $r = 3$,

$$P_\alpha(t) = 1 - 20t + 64t^2 = (1 - 4t)(1 - 16t),$$

homogeneous recurrence $m^3 A_m = (2m - 1)(10(m - 1)^2 + 10(m - 1) + 4)A_{m-1} - 64(m - 1)^3 A_{m-2}$, and $A = 1, 4, 28, 256, 2716, 31504, \dots$. Its level is 12, and:

Computer-assisted finite verification 4.1 (the fold is the Fricke point). [VERIFIED: matches the fold nome to series precision] The fold nome of α is $q_c = e^{-2\pi/\sqrt{12}}$, i.e. $\tau_{12} = i/\sqrt{12}$, the fixed point of the Fricke involution W_{12} on $\mathbb{H}$. Correspondingly $t_c = 1/16$ and the conjugate root is $t'_c = 1/4$.

Now $\dim S_4(\Gamma_0(12)) = 2$: there is no newform, and the space is the oldspace spanned by the two embeddings $f_6(q)$ and $f_6(q^2)$ of the level-6 newform. A two-dimensional space is exactly what Problem 3.8 demands, and the Atkin–Lehner involution splits it into eigenlines.

4.2. The oldspace vanishing lemma. This is the technical heart, and it is proved.

Theorem 4.2 (oldspace swap and vanishing). [PROVED (route complete)] *Let $f_6 \in S_4(\Gamma_0(6))$ be the newform above, with Fricke eigenvalue $\varepsilon_6 := \varepsilon(f_6, W_6) = +1$. Then, on $S_4(\Gamma_0(12)) = \text{span}\{f_6(q), f_6(q^2)\}$, the Fricke involution W_{12} acts by*

$$f_6(q) \mapsto 4f_6(q^2), \quad f_6(q^2) \mapsto \frac{1}{4}f_6(q).$$

Consequently the eigenvectors are $f_6 \mp 4f_6(q^2)$ with eigenvalues ± 1 ; the odd eigenvector

$$f^* := f_6(q) - 4f_6(q^2) = q - 6q^2 - 3q^3 + 12q^4 + 6q^5 + \cdots \in S_4(\Gamma_0(12))$$

satisfies $f^|_4 W_{12} = -f^*$, and*

$$f^*(\tau_{12}) = 0.$$

Proof. The Fricke sign of f_6 is $+1$: from $a_2 = -2$ and $a_3 = -3$ the Atkin–Lehner pseudo-eigenvalues at 2 and 3 are $\lambda_2 = -a_2/2^{k/2-1} \cdot 2^{-1} = 1$ and $\lambda_3 = 1$ in the weight-4 normalization used here, and $\varepsilon_6 = \lambda_2\lambda_3 = +1$.

Compute the slash directly. With $k = 4$,

$$\begin{aligned} (f_6|_4 W_{12})(\tau) &= f_6\left(\frac{-1}{12\tau}\right) (\sqrt{12}\tau)^{-4} = f_6\left(\frac{-1}{6 \cdot (2\tau)}\right) (\sqrt{12}\tau)^{-4} \\ &= \varepsilon_6 (\sqrt{6} \cdot 2\tau)^4 f_6(2\tau) (\sqrt{12}\tau)^{-4} = 4f_6(2\tau), \end{aligned}$$

using $(\sqrt{6} \cdot 2\tau)^4 / (\sqrt{12}\tau)^4 = (24/12)^2 = 4$. Since W_{12} is an involution on the space, the second formula follows, and the eigenvectors and eigenvalues are immediate.

For the vanishing: at the fixed point $\tau_{12} = i/\sqrt{12}$ one has $-1/(12\tau_{12}) = \tau_{12}$ and the weight-4 automorphy factor is $(\sqrt{12}\tau_{12})^4 = (i)^4 = 1$. Hence for the odd eigenvector $f^*(\tau_{12}) = (f^*|_4 W_{12})(\tau_{12}) = -f^*(\tau_{12})$, so $f^*(\tau_{12}) = 0$. $\square$

Remark 4.3 (what this upgraded). Both the relation $f_6(\tau_{12}) = 4f_6(2\tau_{12})$ and the vanishing $f^*(\tau_{12}) = 0$ were first found numerically [VERIFIED: PSLQ, 35⁺ digits]. Theorem 4.2 replaces both by an argument: the constant 4 is not a fitted number, it is the eigenvector condition. This is the single place in the paper where a [VERIFIED:] statement was upgraded, and the upgrade is recorded rather than performed silently. The label is [PROVED (route complete)] rather than [PROVED] only because the normalization of the Atkin–Lehner pseudo-eigenvalues at 2 and 3 is quoted from the standard theory rather than rederived.

The general principle behind Theorem 4.2 is worth stating separately, since it is what a newcomer should carry away.

Corollary 4.4 (odd forms vanish at AL fixed points). [PROVED] *Let $g \in M_k(\Gamma_0(N))$ with $g|_k W_N = -g$ and $k \equiv 0 \pmod{4}$. Then g vanishes at the fixed point $i/\sqrt{N}$ of W_N , because the automorphy factor there is $(\sqrt{N} \cdot i/\sqrt{N})^k = i^k = 1$.*

4.3. The source identity.

Computer-assisted finite verification 4.5 (the level-12 source identity). [VERIFIED: exact coefficientwise to q^{40} ; ambient $M_8(\Gamma_0(12))$, Sturm bound 16]

$$f^* \cdot \sqrt{1-4t} = \Phi_\alpha, \quad \text{equivalently} \quad R^*(t) := \frac{t f^*}{\Phi_\alpha} = \frac{t}{\sqrt{1-4t}}.$$

This identity is the crux, and it says something sharp: P_α factors as $(1-4t)(1-16t)$, and the vanishing combination f^* extracts *only the conjugate root factor* $\sqrt{1-4t}$, dropping $\sqrt{1-16t}$ — the factor that vanishes at the fold $t_c = 1/16$. That is precisely the analytic shadow of $f^*(\tau_{12}) = 0$: the would-be singularity of R^* at the fold has been cancelled by the zero of the source.

The rational reconstruction is exact, and its coefficients are the plainest possible integers:

$$\frac{t}{\sqrt{1-4t}} = \sum_{m \geq 1} \binom{2m-2}{m-1} t^m = t + 2t^2 + 6t^3 + 20t^4 + 70t^5 + \dots$$

— the central binomial coefficients.

4.4. The apparatus.

Construction 4.6 (the manufactured apparatus). Set $B_0^* = 0$, $B_1^* = 1$, and

$$m^3 B_m^* = (2m-1)(10(m-1)^2 + 10(m-1) + 4) B_{m-1}^* - 64(m-1)^3 B_{m-2}^* + \binom{2m-2}{m-1}. \quad (4)$$

Thus

$$B^* = 0, 1, \frac{37}{4}, \frac{818}{9}, \frac{141587}{144}, \frac{51588841}{4500}, \dots$$

and A_n is the Domb sequence $1, 4, 28, 256, 2716, 31504, \dots$

Everything in (4) is over $\mathbb{Z}$: the homogeneous coefficients, the forcing term, the initial data. That is the arithmetic half of the design requirement.

Proposition 4.7. [PROVED] (given Verification 4.5) $B_n^* = [t^n] y^{f^*}$ for the f^* -companion of family α , i.e. (4) is Definition 2.1 for $\Psi = f^*$.

Proof. Immediate from Definition 2.1: the family's homogeneous operator is α 's, and the forcing term is $[t^m] R^* = \binom{2m-2}{m-1}$ by Verification 4.5. $\square$

Computer-assisted finite verification 4.8 (integrality). [VERIFIED: exact, $n \leq 40$] $d_n^3 B_n^* \in \mathbb{Z}$.

We stress the label. This is 40 exact rational computations, not a theorem; the corresponding statement for Apéry's own Eisenstein companion *is* a theorem, by the classical Apéry–Cohen denominator argument, but that argument has not been transported to the cuspidal setting. See §8, item (6).

Computer-assisted finite verification 4.9 (exponential convergence). [VERIFIED: $n \leq 200$ at 120-digit precision] B_n^*/A_n stabilizes at rate $\approx 4^{-n}$; the increment between $n = 100$ and $n = 300$ is $\approx 5 \times 10^{-61}$, and $|B_{200}^*/A_{200} - \xi^*| \approx 10^{-121}$.

Computer-assisted finite verification 4.10 (the limit). [VERIFIED: PSLQ, residual 2×10^{-61} , over the basis $\{\xi^*, L(f_6, 3), L(f_6, 2), \pi L(f_6, 2), \zeta(3), \pi^3, 1\}$ with the stated coefficient cap] With $\xi^* := \lim_n B_n^*/A_n = 0.36455672878339462975728560797\dots$, PSLQ returns the relation $-2\xi^* + L(f_6, 3) = 0$, i.e.

$$\boxed{\lim_{n \rightarrow \infty} \frac{B_n^*}{A_n} = \frac{L(f_6, 3)}{2}}$$

with *no* quasiperiod term, *no* $\pi L(f_6, 2)$, and *no* $\zeta(3)$.

Remark 4.11 (what has been manufactured). Collecting: an integral three-term recurrence with integral forcing, whose two solutions $A_n \in \mathbb{Z}$ and $B_n^* \in d_n^{-3}\mathbb{Z}$ [VERIFIED: $n \leq 40$] satisfy $B_n^*/A_n \rightarrow L(f_6, 3)/2$ with error decaying like 4^{-n} . To our knowledge this is the first Apéry-style apparatus — integral d_n^3 -denominators, exponential convergence, three-term-plus-forcing recurrence over $\mathbb{Z}$ — whose limit is a critical L -value of a cusp form. It was *manufactured, not found*: the level was chosen for $\dim S_4 = 2$, the source was chosen in the Fricke-odd line, and the vanishing at the fold is by construction (Theorem 4.2). This answers Problem 1.1 affirmatively, and answers Problem 3.8 in a stronger form than asked: the second-kind term is not cancelled between two companions, it is never generated.

5. THE MECHANISM: ONE PARITY, TWO EFFECTS

Why did it work? The answer is a single hypothesis — Fricke-oddness of the source at a fold sitting at the involution's fixed point — acting through two logically independent channels.

5.1. Effect 1: fold regularization (exponential convergence).

Lemma 5.1 (fold regularity). [PROVED] (given Verification 4.5) $R^* = t/\sqrt{1-4t}$ is analytic on $|t| < 1/4$; in particular it is regular at the fold $t_c = 1/16$, its nearest singularity being at the conjugate root $t'_c = 1/4$.

The consequence is the convergence rate. The homogeneous solutions grow like $t_c^{-n} = 16^n$; the particular solution generated by an inhomogeneity whose singularity sits at t'_c contributes $(t'_c)^{-n} = 4^n$. Hence

$$\frac{B_n^*}{A_n} - \xi^* \sim \left(\frac{t_c}{t'_c}\right)^n = 4^{-n},$$

matching Verification 4.9. Structurally: a source vanishing at the fold moves the inhomogeneity's singularity off the fold and onto the conjugate root, and the error ratio becomes the root ratio. Contrast the level-6 experiment, where $f_6(\tau_c) \neq 0$ leaves $R = t/\sqrt{P_\gamma}$ singular at the fold itself, so the particular solution has the *same* exponential rate as the homogeneous ones and only a polynomial gap survives (Observation 3.6(2)). We record the general statement:

Proposition 5.2 (regularization criterion). [PROVED (route complete)] For a substituted source Ψ on a rectified family with a simple fold at q_c , the ratio $R = t\Psi/\Phi$ is regular at t_c if and only if Ψ vanishes at q_c to at least the order of the vanishing of Φ there; when it is regular, the approximation error decays geometrically with ratio t_c/t_c^{next} , where t_c^{next} is the nearest singularity of R .

5.2. Effect 2: period parity (a single critical value). The second effect is arithmetic. Let $\Lambda^*(s)$ be the completed L -function of f^* at level 12. Fricke-oddness, $\varepsilon(f^*) = -1$, forces an odd functional equation, and this has an immediate consequence.

Proposition 5.3 (the even critical value dies). [PROVED (route complete)] $\varepsilon(f^*) = -1$ implies

$$\Lambda^*(2) = 0, \quad \Lambda^*(1) = -12\Lambda^*(3),$$

and (by the scaling of the two embeddings) $\Lambda^*(3) = L(f_6, 3)/(8\pi^3)$. All three are confirmed numerically [VERIFIED: to truncation precision].

$\Lambda^*(2) = 0$ is the reason no $L(f_6, 2)$ appears in Verification 4.10: the weight-4 period polynomial of an odd form carries only its odd critical data, and at weight 4 that is a single value. This is the exact contrast with §3, where $\varepsilon(f_6, W_6) = +1$ and both $L(f_6, 3)$ and $L(f_6, 2)$ survive (Verification 3.4).

5.3. The fold split, and where the constant $\frac{1}{2}$ is open. Writing Θ^*, Θ_2^* for the third and second Eichler integrals of f^* , formula (3) at the level-12 fold reads

$$\xi^* = \Theta^*(\tau_{12}) + \frac{2\pi}{\sqrt{12}} \Theta_2^*(\tau_{12}), \quad (5)$$

[VERIFIED: 25 digits], with $F_\alpha|W_{12} = -12\tau^2 F_\alpha$ (an eta computation), so $F'(\tau_{12}) = 12\tau_{12}F(\tau_{12})$. The $\varepsilon = -1$ value identity $\rho^*(\tau_{12}) = 0$ checks, and is equivalent to $\Lambda^*(1) + 12\Lambda^*(3) = 0$ from Proposition 5.3.

Proposition 5.4 (the exact rational factor). [ROUTE WRITTEN, CONSTANT OPEN] *The route to $\xi^* = L(f_6, 3)/2$ is: differentiate the weight-(-2) Eichler cocycle relation for f^* at the fixed point τ_{12} , substitute $\Theta^{*'} = 2\pi i \Theta_2^*$ and $F'/F = 12\tau_{12}$ into (5), and reduce the resulting combination by Proposition 5.3. Every step of this route is written; the hand computation of the derivative fixed-point identity produced an inconsistent orientation/normalization constant, so the exact rational factor $\frac{1}{2}$ is, at the time of writing, established only by Verification 4.10.*

We are explicit about this because it is the paper's one soft joint. The *shape* of the limit — that it is a rational multiple of the single odd critical value $L(f_6, 3)$, with no second-kind contamination — is supported by Propositions 5.3, 5.4 and the split (5). The *value* $\frac{1}{2}$ of the rational factor is at present a 61-digit PSLQ identification.

5.4. Sol's six-item package. For the record, the proof programme that organizes this section, formulated by Sol as a review of the construction, and the status of each item:

item	status
1 source identity $f^*\sqrt{1-4t} = \Phi_\alpha$	[VERIFIED: q^{40}], Sturm-boundable ($M_8(\Gamma_0(12))$, bound 16)
2 forced recurrence (4)	[PROVED] formally, given item 1 (Prop. 4.7)
3 fold-regularity lemma	[PROVED] given $R^* = t/\sqrt{1-4t}$ (Lemma 5.1)
4 Fricke functional equation / oldspace swap	[PROVED (route complete)] (Theorem 4.2)
5 period parity	structure [PROVED (route complete)] (Prop. 5.3); exact constant [ROUTE WRITTEN, CONSTANT OPEN] (Prop. 5.4)
6 denominator theorem $d_n^3 B_n^* \in \mathbb{Z}$	[VERIFIED: $n \leq 40$]; [OPEN] beyond

6. CONTROLS, SCOPE, AND THE GENERAL CONJECTURE

A mechanism claim is only as good as its controls. Two were run.

6.1. The δ control: fold-vanishing is the operative hypothesis. Family δ also lives at level 12, so the *same* source f^* can be substituted into a *different* curve. Sol proposed this as a directed test, and it separates the layers cleanly.

Computer-assisted finite verification 6.1 (the δ control). [VERIFIED: exact, $n \leq 40$; numerics at 120 digits] For the f^* -companion on δ 's curve:

- (1) $d_n^3 B_n^{f^*, \delta} \in \mathbb{Z}$ still holds for $n \leq 40$. The integrality layer is a property of (level, weight, d_n -structure), and is *independent of the curve*.

- (2) But δ 's fold is complex, $q_c^\delta \approx 0.1137 - 0.1970i$, which is *not* the Fricke point, and $f^*(q_c^\delta) \approx 0.292 + 0.103i \neq 0$. There is no regularization: the coefficients of R_δ oscillate with $|\text{ratio}| \rightarrow 9$, both conjugate folds being singular.

So the three design requirements really are carried by three different structures, and the vanishing-*at-the-fold* hypothesis is necessary, not decorative.

6.2. The uniqueness corollary.

Corollary 6.2 (α is the only sporadic host). [PROVED] (given the fold and dimension data of [2, 1]) *A rational vanishing source in the sense of §4 requires (i) a real fold sitting at an Atkin–Lehner fixed point, and (ii) $\dim S_{r+1} \geq 2$ at the family's level. Among the nine order-3 sporadic families, real folds occur only for $\alpha, \gamma, \varepsilon, \zeta$ (levels 12, 6, 8, 9), whose cusp dimensions are 2, 1, 1, 1 respectively. Hence α **is the unique sporadic family admitting the construction.***

Proof. (i) is Corollary 4.4 together with Proposition 5.2: the vanishing must occur at the fold, and the only mechanism producing a forced zero of a rational integral source is oddness under an involution fixing that point; a complex fold is not an Atkin–Lehner fixed point on $\mathbb{H}$ for a real involution. (ii) is needed because at $\dim S_{r+1} = 1$ the eigenline decomposition is trivial and the unique cusp direction has whatever Fricke sign it has — at level 6 it is $+1$, so no odd cuspidal direction exists at all. The dimension and fold data are then read off the table. $\square$

Remark 6.3. The complex-fold families (B, δ, η) are excluded over $\mathbb{R}$ outright by clause (i). In particular Sol's suggested test at level 20 via family η is answered without computation: η 's discriminant is -16 . Further instances therefore require *non-sporadic* rectified families at genus-zero levels with larger cusp spaces — the reverse factory's next search domain, now equipped with a quantitative objective (maximize the error-decay base against the denominator-growth base).

6.3. The general conjecture.

Conjecture 6.4 (D-2, in Sol's refined mechanism form). *For a modular Picard–Fuchs family with a fold fixed by an Atkin–Lehner involution, a rational integral source in the (-1) -eigenspace that vanishes to the required order at the fold yields a regularized companion whose connection value lies in the odd critical-period line. Quantitatively: for every genus-zero level N with $\dim S_{r+1} \geq 2$ and a rectified family whose fold is the W_N -fixed point, the Fricke-odd vanishing combination yields a d_n^r -integral, exponentially convergent apparatus whose limit is a rational multiple of an odd critical L -value.*

Status: [OPEN]. Evidence: two data points (levels 6 and 12, the second positive and the first explaining the necessity of the parity hypothesis), plus the δ control isolating the vanishing clause, plus the three separated layers — *integral structure* $\Rightarrow d_n^r$; *fold-vanishing* $\Rightarrow$ regularization; *odd parity* $\Rightarrow$ odd critical value — each of which is independently supported. Two data points is not evidence for a general law, and we do not present it as such; the value of the conjecture is that it states exactly which hypotheses a third data point must test.

7. HONEST IRRATIONALITY BOOKKEEPING

It would be easy to over-read §4. We therefore state the negative plainly.

7.1. The apparatus loses the race. An Apéry-style irrationality proof needs the cleared error to tend to zero:

$$|d_n^3(A_n \xi^* - B_n^*)| \longrightarrow 0.$$

Here $d_n^3 \sim e^{3n} = (20.08 \dots)^n$ while $A_n |B_n^*/A_n - \xi^*|$ decays only like 4^{-n} relative to A_n ; the product behaves like $e^{3n} \cdot 4^{-n}$ (subexponential) $\rightarrow \infty$.

Observation 7.1 (no irrationality). [PROVED] The apparatus of Construction 4.6 does *not* prove, and does not bear on, the irrationality of $L(f_6, 3)$. The decay base 4 loses to the denominator base $e^3 = 20.08 \dots$

Its value is structural: cuspidal critical L -values are now inside the modular-anchor factory with Apéry-quality arithmetic, and the convergence/denominator trade-off becomes an *optimization problem* over the triple (level, family, source) — the same problem that Apéry’s $\zeta(3)$ construction wins in the Eisenstein class.

7.2. The race, scored over all fifteen families. Sol’s scoring criterion: with $A_n \sim \alpha^n$, error $\sim \beta^n$, and denominators $\ll \delta^n$, one needs $\delta|\beta| < 1$. For our families $|A_n \xi - B_n| \sim |t'_c|^{-n}$ and $\delta = e^r$, so the score is $e^r/|t'_c|$ with t'_c the larger root of P .

Computer-assisted finite verification 7.2 (Route-1 scan). [VERIFIED: exact roots, all fifteen families]

family	r	$ t'_c $	$e^r/ t'_c $	verdict
$D(\zeta(2))$	2	11.09	0.67	win — Apéry’s $\zeta(2)$
$\gamma(\zeta(3))$	3	33.97	0.59	win — Apéry’s $\zeta(3)$
ε	3	1.457	13.8	fail (best of the rest)
all others		≤ 1	≥ 7.4	fail

Exactly the two classical wins, with nothing near the boundary. The sporadic table is exhausted as a source of irrationality proofs.

7.3. Apéry’s miracle is the same construction.

Computer-assisted finite verification 7.3 (the classical source vanishes at its fold). [VERIFIED: $|\Phi_\gamma(\tau_c)| < 10^{-122}$] $\Phi_\gamma(\tau_c) = 0$: Apéry’s own $\zeta(3)$ source vanishes at the fold. It is the W_6 -odd Eisenstein direction, and the vanishing is Corollary 4.4 again — any weight-4 form odd under W_6 vanishes at $i/\sqrt{6}$ since the automorphy factor there is $+1$.

This reframes the whole picture. Apéry’s construction *is* the construction of §4: an odd source vanishing at an Atkin–Lehner-fixed fold. It succeeds where ours does not only because its level has root ratio $34 > e^3$. One mechanism, two classes:

	odd Eisenstein source	odd cusp source
level 6 (ratio $34 > e^3$)	Apéry’s $\zeta(3)$ proof	$S_4^-(\Gamma_0(6)) = 0$ — none exists
level 12 (ratio $4 < e^3$)	$\zeta(3)$ -type, loses race	our $L(f_6, 3)/2$ apparatus, loses race

Note also that the “mixed-source fix” at level 6 — subtract a multiple of the Eisenstein source to force f_6 to vanish at the fold — is impossible: Φ_γ *already* vanishes there, so no multiple of it changes the value of f_6 at τ_c [EXCLUDED: verified].

7.4. The precise open problem.

Problem 7.4 (the cuspidal-Apéry question). [OPEN] Find a genus-zero group Γ carrying a rectified family whose Atkin–Lehner-fixed fold has conjugate root $|t'_c| > e^r$, and with $S_{r+1}^-(\Gamma) \neq 0$. The odd cuspidal companion would then give a race-winning d_n^r -apparatus for a cusp-form critical L -value — an irrationality proof of a cuspidal period — *provided* the d_n^r -integrality, which is proved in the Eisenstein class but only observed in the cuspidal one, survives.

Two failure modes deserve respect. First, cuspidal modular-symbol denominators may be genuinely worse than d_n^r (Sol’s warning); our $n \leq 40$ data cannot see this. Second, the sporadic table itself is exhausted (Verification 7.2, Corollary 6.2), so the search must move to non-sporadic rectified families.

8. EVIDENCE STATUS

Consolidated, with nothing upgraded relative to the ledgers.

- (1) **The Eisenstein/cuspidal dichotomy of sources** (§1): [VERIFIED: exact $\mathbb{Q}$, coefficient-wise to q^{60} , held-out band $q^{27}-q^{60}$], conditional on the modularity certification of t, F per family (complete only for level 9).
- (2) **Level-6 identity** $f_6\sqrt{1-34t+t^2} = \Phi_\gamma$: [VERIFIED: q^{60}], theorem-grade modulo the same certification, via the Sturm bound 8 in $M_8(\Gamma_0(6))$.
- (3) **Level-6 limit** $\Theta(\tau_c) = L(f_6, 3) - \frac{\pi}{\sqrt{6}}L(f_6, 2)$: [VERIFIED: 120 digits]; period-polynomial proof route stated, not written out. Second-kind constant $\Theta_2(\tau_c)$: [OPEN] (PSLQ negative over the stated bases).
- (4) **Oldspace swap and** $f^*(\tau_{12}) = 0$: [PROVED (route complete)] (Theorem 4.2); previously [VERIFIED: PSLQ 35⁺ digits].
- (5) **Source identity** $f^*\sqrt{1-4t} = \Phi_\alpha$: [VERIFIED: exact to q^{40}] via the central-binomial reconstruction of R^* ; Sturm-boundable in $M_8(\Gamma_0(12))$ (bound 16), which the verification range exceeds.
- (6) **d_n^3 -integrality** of B^f and B^* : [VERIFIED: $n \leq 40$] in both cases; [OPEN] beyond. No denominator theorem is claimed.
- (7) **The limit** $B_n^*/A_n = L(f_6, 3)/2$: [VERIFIED: 61-digit PSLQ, residual 2×10^{-61}], with 121-digit stability of the ratio; the shape (rational multiple of the odd critical value) is supported by Propositions 5.3–5.4, the factor $\frac{1}{2}$ is [ROUTE WRITTEN, CONSTANT OPEN].
- (8) **Mechanism**: Effect 1 [PROVED] given item (5) (Lemma 5.1); Effect 2 structurally [PROVED (route complete)] (Proposition 5.3), exact constant [ROUTE WRITTEN, CONSTANT OPEN].
- (9) **δ control**: [VERIFIED: $n \leq 40$ exact, 120-digit numerics]. **Uniqueness of α** : [PROVED] given the tabulated fold and dimension data.
- (10) **κ -cancellation**: [EXCLUDED: PSLQ, 40 digits, stated bases and height caps].
- (11) **Race scan and** $\Phi_\gamma(\tau_c) = 0$: [VERIFIED: exact roots; 10^{-122}].
- (12) **Conjecture D-2**: [OPEN], two data points.
- (13) **Irrationality**: none. Observation 7.1 is a proved negative about *this* apparatus, not about $L(f_6, 3)$.

Remark 8.1 (for the newcomer). The pattern to internalize: a 61-digit PSLQ identification is overwhelming *evidence* and zero *proof*; a q^{40} coefficient agreement inside a space with Sturm bound 16 *is* a proof, once the ambient space is certified; and an integrality check to $n = 40$ is neither, because denominator theorems in this subject fail first at moderate n for reasons invisible in small data. Keeping these three apart is the whole of the discipline.

9. PROVENANCE AND DATA

All computations are logged in the programme's ledgers, with exact arithmetic (Python `Fraction`) for series work and `mpmath` at 120 decimal digits for the numerics.

- Source dichotomy and the twelve identifications: `work/PHI_SOURCE_LEDGER.md`; scripts `work/z5eps/eps60_phi_source.py`, `eps60b_phi_holdouts.py`, `eps61_eichler_limits.py`.
- Level-6 experiment (§3): `work/CUSPIDAL_COMPANION.md`; scripts `work/z5eps/eps62_cuspidal_companion.py`, `eps62b_cusp_limit.py`.
- Level-12 construction, δ control, uniqueness, proof notes, race scan (§4–§7): `work/D1_CUSPIDAL_APPARATUS.md`, §§1–9.
- Proved base (rectification, companion inversion, uniform companion formula): `papers_out/modular_anchors/main.tex`.

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